Sakthi Dental Clinic — MERN Website Project Plan

Oct 23, 2026 · @Mahajan Sarthak Dileep

Step-by-step build instructions for a coding agent live in Implementation Runbook.

1. Understanding the project

We are building a 7-page, mobile-first marketing website for Sakthi Dental Clinic, Hosur, on the MERN stack, delivered in 10 working days. The site's job is to make a first-time visitor trust the clinic within seconds and then either press **Fix an Appointment**, call, or send a message.

Who the client is. A Hosur clinic founded in 2004 by Dr. Anupriya, known for dental care for women and children, offering everything from consultations to implants and orthodontics. The brand must read as professional, warm and credible.

Who the users are. Patients of all ages, parents and guardians booking for children, and senior citizens. Most arrive on a phone. This drives three non-negotiables: large readable text, obvious navigation, and a CTA that is never more than one tap away.

What the site must do.

- Introduce the clinic's specialization and build trust (doctor credentials, 20+ years, 9-doctor team, testimonials).
- Showcase treatments: 6 on Home, all 15 on a dedicated Treatments page.
- Show amenities: parking, wheelchair access, pickup and drop-off, central location, daily doctor availability.
- Answer 14 common questions on an FAQs page.
- Capture enquiries through a validated contact form, stored in MongoDB.
- Publish the Privacy Policy verbatim.

Scope boundaries (mandatory from the brief).

- **CTA buttons only.** "Fix an Appointment" and the optional "Emergency Dental Support" are built as styled buttons with a single integration hook. The ShadowFox web development team wires their action, so we build **no booking system**.
- Only copyright-free images and icons, plus the assets in the client's Drive folder.

- Content is used as supplied; the Privacy Policy is copied exactly.
- Out of scope: online booking, payments, patient login, admin dashboard, blog, multilingual content. These are listed so nobody silently adds them inside a 10-day window.

Why MERN fits. MongoDB holds clinic content (treatments, doctors, FAQs, testimonials, amenities, clinic info) and contact submissions. Express exposes a small versioned REST API. React renders the pages, prerendered at build time so search engines see full HTML. Node runs the API. Content changes later need a database edit, not a code release.

2. Requirements traceability matrix

All 52 requirements in the brief map to a build location and a check; the final review (section 19) walks this table row by row.

ID	Requirement from the brief	Where it is built	Verified by
G1	Copyright-free images and icons only	`/client/public/images`, lucide-react icons, Drive assets; `CREDITS.md` lists every source and licence	Asset audit before release
G2	Reflect professionalism, warmth, credibility	Design system (section 6), trust strip, doctor credentials	Coordinator design review
G3	Diverse audience incl. seniors: clarity, easy navigation	18 px body text, 48 px tap targets, max 5 nav items, plain labels	Accessibility checklist (section 13)
G4	"Fix an Appointment" CTA prominent; button only	`AppointmentButton` in header, hero, mobile sticky bar, section ends	E2E: button visible on every page
G5	Design for mobile first	Mobile-first Tailwind breakpoints from 360 px	Playwright mobile viewports
C1	Clean, modern, minimalistic design	Whitespace scale, 2 typefaces, soft palette	Design review
C2	Smooth content flow, clear navigation	Section order per page (section 5), sticky header, anchor links	Usability walk-through

ID	Requirement from the brief	Where it is built	Verified by
C3	Emphasis on Treatments, Amenities, Testimonials, Contact	Dedicated Home sections + Treatments and Contact pages	Content checklist
C4	Highlight doctor credentials and clinic specialties	About page founder bio, 9-doctor grid with specialties, Why Choose cards	Content checklist
F1	Prominent hero with specialization in women and children	`Hero` component, Home	Visual review
F2	Quick nav: About Us, Treatments, FAQs, Contact	`Navbar` (desktop) + `MobileMenu` drawer	E2E nav test
F3	Visually engaging treatments showcase	`TreatmentCard` grid, icons + images	Visual review
F4	Amenities: parking, wheelchair access, transport	`Amenities` section, Home	Content checklist
F5	Testimonials in attractive cards	`TestimonialCarousel`	Visual review
F6	Footer: contact details, nav shortcuts, services list	`Footer` on every page	E2E footer test
H1	Hero headline, subtext, Fix an Appointment, optional Emergency CTA	`Hero` with exact copy; `EmergencyButton` included	Copy proofread
H2	Assurance banner "You are always in safe hands..." with Drive banner images	`AssuranceBanner`	Visual review
H3	Why Choose: 4 points with exact copy	`WhyChoose` 4 cards	Copy proofread
H4	Explore Our Services: 2 or 3 columns, icons/images, 6 treatments	`CoreTreatments` 1/2/3 columns by breakpoint	Responsive test

ID	Requirement from the brief	Where it is built	Verified by
H5	CTA "View Full List of Treatments" redirects to Treatments page	Link to `/treatments`	E2E link test
H6	"What Our Patients Say", rotating quotes, 3 testimonials	`TestimonialCarousel` with auto-rotate + pause	E2E + axe
H7	Clinic Facilities: 5 items with small icons	`Amenities` icon list	Content checklist
H8	Footer: address/contact, Home, About, Treatments, Privacy Policy, key treatments, Instagram, Facebook, YouTube	`Footer`	E2E footer test
A1	"Get to Know Dr. Anupriya" + subtitle + 4 bio paragraphs	`/about` founder section	Copy proofread
A2	Our Vision & Mission	`/about` section	Copy proofread
A3	Team of 9 doctors with roles	`/about` `DoctorGrid`	Count = 9
T1	Separate Treatments page with all 15 treatments	`/treatments`, anchor per treatment	Count = 15
I1	Use Drive images (banners, icons)	Downloaded, optimized to WebP/AVIF	Asset audit
Q1	14 FAQs	`/faqs` accordion	Count = 14
K1	Contact title "We would love to hear from you", form on left	`/contact` two-column layout	Visual review
K2	Fields: Name, Email, Phone (required), Message (optional)	`ContactForm`	Unit + E2E
K3	Validation for required fields	zod schema shared by client and server	Unit + API tests
K4	Email format verification	zod `email()` client and server	Unit + API tests

ID	Requirement from the brief	Where it is built	Verified by
K5	Stacked fields on mobile	Single-column form under 768 px	Mobile viewport test
K6	Submit button labelled "Submit", primary accent colour	`Button variant="primary"`	Visual review
K7	"Reach us" on right: address + pin, email + mail, phone + phone, timings + clock, timing bold	`ReachUs` with lucide icons	Visual review
K8	Contact page clean, professional, fully responsive	Layout + tests	Responsive test
P1	Privacy Policy copied exactly	`/privacy-policy`, text stored verbatim	Diff against brief
N1	Mobile responsive, cross-browser	Breakpoints + browserslist	Playwright Chromium, Firefox, WebKit
N2	SEO-optimized structure, clean code	Prerender, meta, JSON-LD, sitemap; ESLint + Prettier	Lighthouse SEO 100
N3	Fast-loading assets, optimized images	sharp pipeline, lazy loading, code splitting	Lighthouse performance 90+
V1	Professional but friendly tone	Copy + illustration style	Design review
V2	Trust-building and informative	Credentials, FAQs, testimonials	Design review
V3	Whitespace, soft lavender/white/pastel tones, clean typography	Design tokens (section 6)	Design review
M1	Finish in 7 to 10 days	10-day plan (section 18)	Daily coordinator update

ID	Requirement from the brief	Where it is built	Verified by
M2	Split work into UI mockups, responsiveness, interactivity, testing	Timeline phases	Timeline review
M3	Update coordinator regularly	End-of-day update template	Sent daily
S1	MERN stack (your condition)	MongoDB, Express, React, Node (section 7)	Architecture review
S2	Contact submissions persisted	`contact_messages` collection	API test
S3	Content served from database	Content collections + seed script	API test
S4	No booking logic built (dev team owns CTA action)	Single `onAction` hook, no-op by default	Code review
S5	Privacy Policy claims stay true (no selling data, cookies disclosed)	No third-party trackers by default	Network audit

3. Content gaps and conflicts to confirm

The brief has 10 gaps or contradictions; send this list to the coordinator on Day 1 and build with the default shown until they answer.

#	Issue	What the brief says	Default until confirmed
1	Hours conflict	Amenities: doctors available daily 9 AM to 9 PM. Contact: Sunday to Saturday 9am to 7pm	Show both exactly as written; store both in `clinic_info` so one edit fixes it
2	Doctor name typo	"Dr. Ajay Jumar (Oral & Maxillofacial Surgeon)"	Keep as given; ask if it should be "Kumar"
3	Founder timeline	Graduated 2000, six years at Mathura Clinic, founded clinic in 2004	Keep verbatim; flag the overlap

#	Issue	What the brief says	Default until confirmed
4	Doctor credentials	Brief asks to highlight credentials but gives only names and specialties	Show name + specialty; ask for degrees (BDS/MDS), registration numbers and photos. Never invent them
5	Social media links	Instagram, Facebook, YouTube icons requested, no URLs given	Icons link to `#` with `aria-disabled`; URLs live in `clinic_info.socials`
6	Drive assets	Banner images and icons are in a Google Drive folder	Download on Day 1; if access fails, use Unsplash/Pexels placeholders and swap later
7	Treatment naming	Home lists "Bleaching", "Orthodontic Treatment", "Artificial Complete Denture"; Treatments page uses "Teeth Whitening (Bleaching)", "Orthodontic Braces"	Store `homeTitle` and `title` separately; Home card links to the matching anchor
8	Testimonial ratings	Only Manisha M has a rating (5/5)	Show stars only for her; no invented ratings
9	Accent colour	Submit button "blue/pink"	Pink accent for CTAs (section 6); one token, easy to switch
10	Privacy Policy law reference	Policy cites COPPA (US); clinic is in India	Copy exactly as instructed; mention to coordinator that India's DPDP Act may be more relevant

Two small text fixes are safe without asking: the stray line break inside the Teeth Whitening paragraph ("several / shades"), and inconsistent testimonial name formatting ("Mr. *Arun Kumar*").

4. Sitemap and navigation

The site has 6 content routes plus a 404, all one level deep so no page is more than one tap from the header.

Route	Page	Header nav	Footer quick links	Prerendered
`/`	Home	Logo link	Home	Yes
`/about`	About Us	About Us	About	Yes
`/treatments`	Treatments (all 14, anchor per item, e.g. `/treatments#root-canal-therapy`)	Treatments	Treatments	Yes
`/faqs`	FAQs	FAQs	-	Yes
`/contact`	Contact Us	Contact	-	Yes
`/privacy-policy`	Privacy Policy	-	Privacy Policy	Yes
`*`	404 with links home and to Contact	-	-	Static

Header (desktop, 1024 px and up). Logo left; About Us, Treatments, FAQs, Contact centre; "Fix an Appointment" button right. Sticky, shrinks from 80 px to 64 px on scroll, active link underlined.

Header (mobile). Logo, a call icon (`tel:+919862890897`) and a hamburger. The drawer lists the same four links at 20 px with 56 px rows and closes on Escape or route change.

Mobile sticky action bar. Under 768 px, a bottom bar holds "Fix an Appointment" and "Call". It hides on `/contact` while the form is focused so it never covers the keyboard.

Footer (every page). Four columns that stack on mobile: clinic address and contact; quick links (Home, About, Treatments, Privacy Policy); key treatments (the 6 core ones, each linking to its anchor); social icons (Instagram, Facebook, YouTube). Bottom row: copyright line with the current year.

Breadcrumbs. Not needed for a one-level site; each page shows a page-title banner instead.

5. Page-by-page blueprint

Each page below lists its sections top to bottom, with layout per breakpoint; all copy comes from the brief via the seed data, never hard-coded in components.

Home `/`

1. **Hero** (`Hero`). Left: eyebrow "Sakthi Dental Clinic, Hosur", H1 "Specialized Dental Care for Women, Children & Families", subtext as given, two buttons: "Fix an Appointment" (primary) and "Emergency Dental Support" (outline). Right: warm photo of a dentist with a mother and child. Under it, a trust strip: "Since 2004", "20+ years of expertise", "9 specialist doctors", "Open daily". Mobile: text first, image below at 4:3.
2. **Assurance banner** (`AssuranceBanner`). Full-width Drive banner image with a lavender overlay and the line "You are always in safe hands. We are ready to help, anytime." plus a call link. Height 280 px desktop, 200 px mobile.
3. **Why Choose Sakthi Dental Clinic?** (`WhyChoose`). One intro sentence on the clinic's philosophy, then 4 icon cards: All-in-One Care, Experienced Doctors, Patient-Centric Approach, Technology-Driven Services. 4 columns desktop, 2 tablet, 1 mobile.
4. **Explore Our Services** (`CoreTreatments`). 6 cards (Tooth Extraction, Artificial Complete Denture, Tooth Filling, Teeth Cleaning or Scaling, Bleaching, Orthodontic Treatment), each with icon, title, one-line description and "Learn more" to its anchor. 3 columns desktop, 2 tablet, 1 mobile. Centered button "View Full List of Treatments" to `/treatments`.
5. **What Our Patients Say** (`TestimonialCarousel`). 3 quote cards with name; star row only for Manisha M. Desktop shows all 3 side by side; tablet and mobile show 1 card that auto-rotates every 6 s, with dots, previous/next buttons and a pause button. Auto-rotate stops on hover, focus, or `prefers-reduced-motion`.
6. **Clinic Facilities** (`Amenities`). 5 items with icons: Convenient central location (MapPin), Hassle-free parking (ParkingSquare), Doctors available daily 9 AM to 9 PM (Clock), Pickup & drop-off support (Car), Wheelchair access (Accessibility). 5 across desktop, 2 columns mobile.
7. **Closing CTA band** (`CtaBand`). "Ready for a healthier smile?" with Fix an Appointment and Call buttons.
8. **Footer**.

About Us `/about`

1. **Page banner**: "About Us".

2. **Get to Know Dr. Anupriya** (`FounderProfile`). Photo left, text right: subtitle "Your Trusted Partner in Dental Care" and the 4 paragraphs as given. A credential panel beside the photo: Graduated 2000; 6 years at Mathura Clinic; Founded Sakthi Dental Clinic, 2004; Dental consultant, Primary Health Center at Chandara Hospital, 10+ years.
3. **Our Vision & Mission** (`VisionMission`). Two cards side by side (stacked on mobile), Mission first as in the brief.
4. **Our Team of Doctors** (`DoctorGrid`). 9 cards: avatar (neutral placeholder until photos arrive), name, specialty; founder card highlighted. 3 columns desktop, 2 tablet, 1 mobile.
5. **CTA band** + Footer.

Treatments `/treatments`

1. **Page banner**: "Our Treatments" + one-line intro.
2. **Jump list** (`TreatmentIndex`). 15 chips linking to anchors; horizontally scrollable on mobile.
3. **Treatment sections** (`TreatmentDetail` x 15), in brief order: Teeth Cleaning & Scaling, Tooth Filling, Tooth Extraction, Artificial Complete Denture, Dental Implants, Laser Dentistry, Root Canal Therapy, Wisdom Tooth Extraction, Fixed Partial Denture (Bridge), Teeth Whitening (Bleaching), Veneers, Pediatric Dentistry, Flap Surgery, Orthodontic Braces, Clear Aligners. Each: image, H2 title, full description, small "Fix an Appointment" button. Image side alternates on desktop; image on top on mobile. Each section has `id` = slug and `scroll-margin-top` so the sticky header never hides the heading.
4. **CTA band** + Footer.

Order and wording follow the brief exactly. Home card titles map to these anchors (section 3, item 7).

FAQs `/faqs`

1. **Page banner**: "Frequently Asked Questions".
2. **Accordion** (`FaqAccordion`). 14 questions in brief order; one open at a time; first item open by default. Each header is a `<button>` with `aria-expanded` .
3. **Still have questions?** card linking to `/contact` + Footer.

Contact Us `/contact`

1. **Page banner**: "Contact Us".
2. **Two columns** (stacked on mobile, form first). Left: "We would love to hear from you" + `ContactForm` (section 12). Right: "Reach us" (`ReachUs`): address with MapPin, email with Mail (`mailto:`), both phones with Phone (`tel:` links), timings with Clock and the hours in bold.
3. **Map link**: a static map image linking to Google Maps directions (no embedded iframe, to keep the page fast).
4. Footer.

Privacy Policy `/privacy-policy`

Policy text exactly as supplied, sections 1 to 9, in a 720 px reading column with a "Last updated" line. Footer.

404

Friendly message, links to Home and Contact, `noindex` meta.

6. Design system

A soft lavender-and-white base with one pink accent reserved for action buttons, so "Fix an Appointment" is always the most visible thing on screen. All values live as Tailwind v4 `@theme` tokens in `client/src/styles/tokens.css` .

Colour tokens

Token	Hex	Use	Contrast on white
`lavender-50`	#F6F4FD	Alternate section background	Background only
`lavender-100`	#ECE8FB	Cards, banner tint, input fill	Background only
`lavender-300`	#C9BFF2	Borders, decorative shapes	Decorative only
`lavender-700`	#5B4BB7	Headings accent, links, icons, focus ring	6.7:1 (AA)

Token	Hex	Use	Contrast on white
`rose-600` (accent)	#B83280	Primary CTA and Submit background	White text on it 5.5:1 (AA)
`rose-50`	#FDF0F6	Emergency button hover, highlight chips	Background only
`ink-900`	#1F1B2E	Body and heading text	16:1 (AAA)
`ink-600`	#55506A	Secondary text	7.7:1 (AAA)
`success-600` / `error-600`	#1E7A55 / #B42318	Form states	AA

Typography

- Headings: **Plus Jakarta Sans** 600/700; body: **Inter** 400/500. Both are SIL Open Font License, self-hosted through `@fontsource`, Latin subset, `font-display: swap`.
- Fluid scale with `clamp()`: H1 36 to 56 px, H2 28 to 40 px, H3 20 to 24 px, body **18 px** (never below 16 px), small 15 px.
- Line height 1.6 for body, 1.2 for headings; reading width capped at 70 characters.

Layout and spacing

- 8 px spacing grid; section padding 96 px desktop, 64 px mobile.
- Container max width 1200 px; gutters 24 px (16 px under 400 px).
- Breakpoints: 360 base, `sm` 640, `md` 768, `lg` 1024, `xl` 1280.
- Radius 16 px for cards, 999 px for buttons; one soft shadow `0 8px 24px rgb(91 75 183 / 0.08)`.

Core components

- `Button`: variants `primary` (rose), `secondary` (lavender outline), `ghost`; sizes `md` 48 px and `lg` 56 px tall.
- `AppointmentButton`, `EmergencyButton`: wrappers over `Button` (section 10).
- `SectionHeading` (eyebrow + title + intro), `Card`, `IconBadge`, `TreatmentCard`, `TestimonialCard`, `DoctorCard`, `Accordion`, `FormField`, `PageBanner`, `CtaBand`.

Icons and imagery

- Icons: **lucide-react** (ISC licence) at 24 px in a 48 px lavender circle, plus the Drive icon set where it fits.
- Photos: Drive banners first, then Unsplash or Pexels (free licence). Bright clinic interiors, women and children patients, natural smiles; no blood, needles or close-up procedures.
- Each image gets a 4:3 or 16:9 frame, a lavender low-quality placeholder, and descriptive `alt` text.

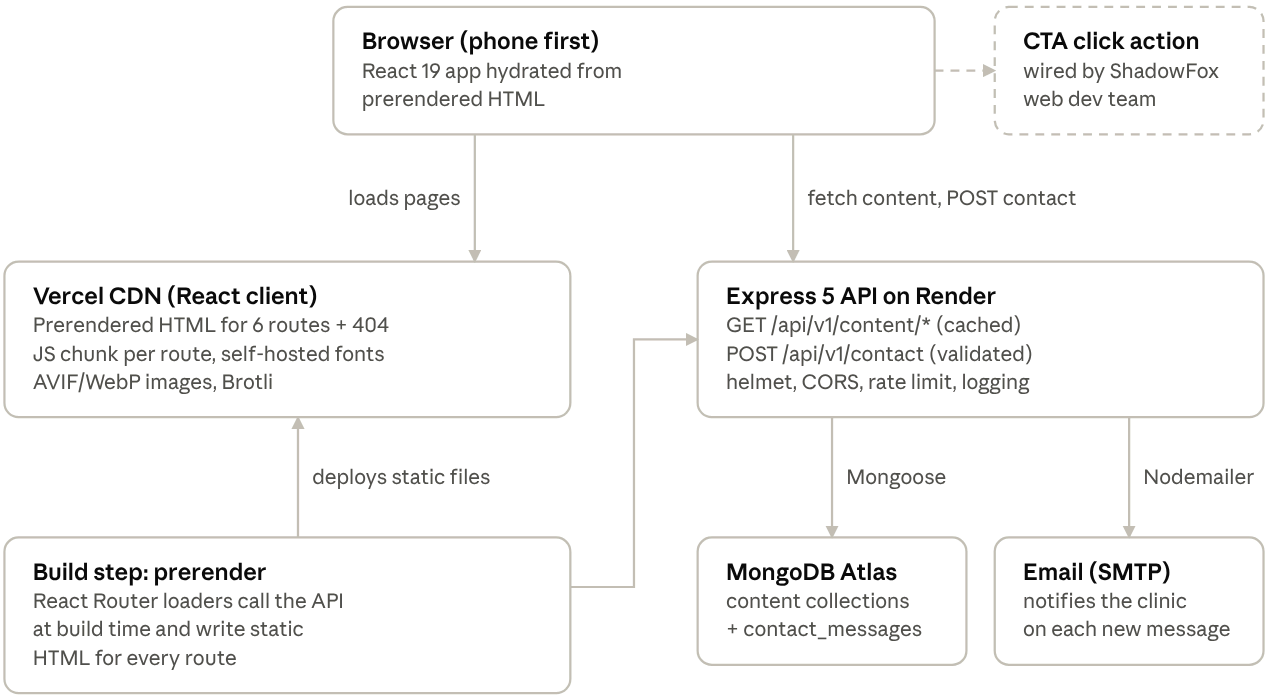
Motion

Fade-and-rise on scroll (200 ms, 16 px) via `IntersectionObserver` and CSS, no animation library. Hover lifts cards 4 px. Everything is disabled under `prefers-reduced-motion: reduce`.

7. MERN architecture and tech stack

React pages are prerendered at build time and served from a CDN; a single Express API reads and writes MongoDB and emails the clinic when someone submits the contact form.

Prerendered React on a CDN, one Express API, one MongoDB



system architecture · client, API, database, email

Patients get full HTML instantly from the CDN, so the site is fast and search engines see every word. The API is only hit for the contact form and for content refreshes after hydration.

Layer	Choice	Why
Runtime	Node.js 24 LTS	Current LTS line, long support window
Frontend	React 19 + React Router v7 (framework mode, `prerender` on) + Vite	Meets the SEO requirement with static HTML per route while staying pure React
Styling	Tailwind CSS v4 with `@theme` tokens	Fast mobile-first layout, tokens map 1:1 to section 6
Forms	react-hook-form + zod	Light, accessible errors, same schema as the server
Icons	lucide-react	Free ISC licence, tree-shakable
API	Express 5	Native async error handling, minimal

Layer	Choice	Why
Database	MongoDB Atlas (free M0 tier to start) + Mongoose 8	The M in MERN; schemas and indexes in code
Email	Nodemailer over SMTP	Notifies `info@sakthidentalclinic.in` on each submission
Shared code	`shared/` workspace with zod schemas and constants	One validation source for client and server
Testing	Vitest, React Testing Library, Supertest, mongodb-memory-server, Playwright, axe-core	Unit, API, end-to-end and accessibility coverage
Tooling	npm workspaces, ESLint, Prettier, Husky + lint-staged	Clean code (N2) enforced on every commit
Hosting	Vercel (client), Render (API), Atlas (DB)	Free tiers cover a clinic site's traffic
CI	GitHub Actions + Lighthouse CI	Lint, test, build and performance gates on every PR

Fallback if framework mode feels heavy: a plain Vite SPA with `react-helmet-async` plus a small Puppeteer script that visits each route after build and saves the HTML. Same output, more manual wiring.

8. MongoDB data model

Eight collections: seven hold clinic content seeded from the brief, one stores contact submissions. Every content document has `order` (for display sequence) and `isActive` (to hide without deleting), plus Mongoose `timestamps`.

Collection	Key fields	Indexes	Seeded count
`clinic_info` (singleton)	`name`, `tagline`, `address{line1, area, city, state, pincode, country}`, `emails[]`, `phones[]`, `timings` ("Sunday to Saturday: 9am to 7pm"), `doctorAvailability` ("Doctors available daily (9 AM - 9 PM)"), `socials{instagram, facebook, youtube}`, `mapUrl`, `hero{headline, subtext}`, `assurance`, `privacyPolicy{markdown, updatedAt}`	`_id` only	1
`treatments`	`title`, `slug`, `homeTitle`, `homeSummary`, `description`, `icon`, `image{src, alt}`, `isCore` (shown on Home), `order`	`slug` unique; `{isCore, order}`	15 (6 with `isCore: true`)
`doctors`	`name`, `specialty`, `isFounder`, `qualifications` (empty until confirmed), `photo{src, alt}`, `order`	`order`	9
`testimonials`	`quote`, `author`, `rating` (nullable), `order`	`order`	3
`faqs`	`question`, `answer`, `order`	`order`	14
`amenities` + `why_choose`	`title`, `description` (why_choose only), `icon`, `order`	`order`	5 + 4

Collection	Key fields	Indexes	Seeded count
`contact_messages`	`name`, `email`, `phone`, `message`, `status` (`new`, `read`, `replied`), `source` (`contact-page`), `userAgent`, `ipHash` (SHA-256, not the raw IP), `emailNotified`	`createdAt` desc; TTL on `createdAt` (365 days, pending coordinator approval)	0

Seeding. `server/src/seed/content.json` holds every piece of copy from the brief, typed once. `npm run seed` validates it with the shared zod schemas, then upserts by `slug` or `order`, so re-running is safe. The Privacy Policy is stored as Markdown exactly as supplied.

Example document (treatments).

```
{
  "title": "Teeth Whitening (Bleaching)",
  "slug": "teeth-whitening",
  "homeTitle": "Bleaching",
  "homeSummary": "Cosmetic whitening treatments for a brighter smile.",
  "description": "Brighten your smile with professional teeth whitening
services...",
  "icon": "sparkles",
  "image": { "src": "/images/treatments/teeth-whitening.webp", "alt":
"Smiling woman after teeth whitening" },
  "isCore": true,
  "order": 10,
  "isActive": true
}
```

9. REST API specification

The API has 9 read endpoints and 1 write endpoint under `/api/v1`, all returning JSON in one envelope: `{ "success": true, "data": ... }` or `{ "success": false, "error": { "code", "message", "fields"? } }`.

Method	Path	Purpose	Response / notes	Cache
GET	`/api/v1/health`	Uptime and DB ping	`{ status, db, uptime }`	none
GET	`/api/v1/content/home`	Everything Home needs in one call	`clinic` (hero, assurance), `whyChoose[]`, `coreTreatments[]` (6), `testimonials[]`, `amenities[]`	10 min
GET	`/api/v1/clinic`	Address, phones, emails, timings, socials	`clinic_info` minus `privacyPolicy`	10 min
GET	`/api/v1/treatments`	All active treatments	Sorted by `order`; `?core=true` returns the 6 Home items	10 min
GET	`/api/v1/treatments/:slug`	One treatment (reserved for future detail pages)	404 `NOT_FOUND` if missing	10 min
GET	`/api/v1/doctors`	Team of 9	Sorted by `order`, founder first	10 min
GET	`/api/v1/testimonials`	Patient quotes	Sorted by `order`	10 min
GET	`/api/v1/faqs`	14 FAQs	Sorted by `order`	10 min
GET	`/api/v1/privacy-policy`	Policy Markdown + last updated	Verbatim text	1 day
POST	`/api/v1/contact`	Save an enquiry, email the clinic	201 `{ id }`; 422 `VALIDATION_ERROR` with `fields`; 429 `RATE_LIMITED`; 500 `SERVER_ERROR`	never

POST /api/v1/contact body.

```
{
  "name": "Revathi S",
  "email": "revathi@example.com",
  "phone": "+91 98765 43210",
  "message": "Is pediatric check-up available on Sunday?",
  "website": ""
}
```

`website` is a hidden honeypot: if it is filled, the API returns 201 but stores nothing. Read endpoints send `Cache-Control: public, max-age=600, stale-while-revalidate=86400`. No endpoint exposes contact messages; reading them is done in Atlas until an admin view is approved.

10. Frontend architecture

One monorepo with three npm workspaces; the client is organized by route, with shared UI in `components/` and every button that needs the dev team's action going through one file.

```
sakthi-dental-clinic/
├── client/                          React Router v7 + Vite
│   ├── app/
│   │   ├── root.tsx                 <html>, fonts, Header, Footer,
MobileActionBar
│   │   ├── routes.ts                route table (section 4)
│   │   ├── routes/
│   │   │   ├── home.tsx             loader -> GET /content/home
│   │   │   ├── about.tsx            loader -> GET /doctors
│   │   │   ├── treatments.tsx       loader -> GET /treatments
│   │   │   ├── faqs.tsx             loader -> GET /faqs
│   │   │   ├── contact.tsx          loader -> GET /clinic
│   │   │   ├── privacy-policy.tsx   loader -> GET /privacy-policy
│   │   │   └── not-found.tsx
│   │   ├── components/
│   │   │   ├── layout/              Header, Navbar, MobileMenu,
MobileActionBar, Footer, PageBanner
│   │   │   ├── cta/                 AppointmentButton, EmergencyButton,
CallButton, ctaConfig.ts
```

```
|   |   |   ├── home/               Hero, AssuranceBanner, WhyChoose,
CoreTreatments,
|   |   |   |                       TestimonialCarousel, Amenities, CtaBand
|   |   |   ├── about/              FounderProfile, VisionMission,
DoctorGrid, DoctorCard
|   |   |   ├── treatments/         TreatmentIndex, TreatmentDetail,
TreatmentCard
|   |   |   ├── faqs/               FaqAccordion
|   |   |   ├── contact/            ContactForm, ReachUs
|   |   |   └── ui/                 Button, Card, IconBadge, SectionHeading,
FormField, Seo
|   |   ├── lib/                    api.ts (fetch wrapper), seo.ts, jsonLd.ts
|   |   └── styles/                 tokens.css, globals.css
|   ├── public/                     images/, icons/, favicon, robots.txt, og-
image.jpg
|   └── react-router.config.ts      prerender: ["/", "/about", "/treatments",
"/faqs", "/contact", "/privacy-policy"]
├── server/                         Express API (section 11)
├── shared/                         zod schemas, constants (phone regex,
limits)
├── e2e/                            Playwright specs
└── package.json                    workspaces + root scripts
```

Data flow. Each route's `loader` calls the API through `lib/api.ts`. At build time the loaders run once and the HTML is saved; in the browser they run again on navigation. If the API is down during in-app navigation, each route's `ErrorBoundary` shows a friendly message with the clinic phone number; a direct visit or reload still gets the prerendered HTML from the CDN, so a patient never sees a blank page.

CTA integration hook (scope boundary S4). Every "Fix an Appointment" and "Emergency Dental Support" button in the site renders through these components, so the dev team wires the action in exactly one place:

```
// client/app/components/cta/ctaConfig.ts
export const ctaHandlers = {
  appointment: () => {},   // ShadowFox web dev team: replace with booking
action
  emergency: () => {},     // ShadowFox web dev team: replace with emergency
action
};

// client/app/components/cta/AppointmentButton.tsx
```

```
export function AppointmentButton({ size = "md", location }: Props) {
  return (
    <Button
      variant="primary"
      size={size}
      data-cta="fix-appointment"
      data-cta-location={location}   // "header" | "hero" | "mobile-bar" |
"treatment:<slug>"
      onClick={ctaHandlers.appointment}
    >
      Fix an Appointment
    </Button>
  );
}
```

The `data-cta` and `data-cta-location` attributes also let the dev team attach analytics or a modal without touching component code.

State. No global store is needed: content comes from loaders, form state lives in react-hook-form, and UI state (menu open, carousel index, open FAQ) stays local with `useState`.

11. Backend architecture

The Express server is layered routes → controllers → services → Mongoose models, with validation and errors handled in middleware so each controller stays under 20 lines.

```
server/src/
├── app.js              builds the Express app (used by tests and index.js)
├── index.js            connects MongoDB, starts the server, handles SIGTERM
├── config/env.js       reads and validates env vars with zod at startup
├── routes/             content.routes.js, contact.routes.js,
health.routes.js
├── controllers/        thin: read req, call service, send envelope
├── services/           content.service.js, contact.service.js,
mail.service.js
├── models/             ClinicInfo, Treatment, Doctor, Testimonial, Faq,
Amenity, WhyChoose, ContactMessage
├── middleware/         validate(schema), rateLimit, errorHandler, notFound,
requestId
├── seed/               content.json, seed.js
└── utils/              logger.js (pino), hashIp.js
```

Middleware order. `requestId` → `pino-http` logging → `helmet` → `cors` (allow-list: production domain, Vercel previews, localhost) → `compression` → `express.json({ limit: "10kb" })` → routes → `notFound` → `errorHandler`.

Contact flow.

1. `rateLimit` allows 5 submissions per IP per 15 minutes on `POST /contact`.
2. `validate(contactSchema)` runs the shared zod schema; failures return 422 with per-field messages.
3. Honeypot check: a filled `website` field returns 201 and stops.
4. `contact.service` trims and strips HTML from all fields, hashes the IP, saves the document.
5. `mail.service` sends a plain-text email to `info@sakthidentalclinic.in` with name, phone, email and message. A mail failure is logged and `emailNotified: false` is saved, but the patient still gets success because the message is safely stored.
6. Controller returns 201 `{ id }`.

Errors. One `AppError(code, status, message)` class. The error handler maps zod errors to 422, Mongoose `CastError` to 400, unknown errors to 500, and never leaks stack traces in production.

Environment variables. `NODE_ENV`, `PORT`, `MONGODB_URI`, `CORS_ORIGINS`, `SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASS`, `MAIL_TO`, `IP_HASH_SALT`. A committed `.env.example` lists them with no secrets.

12. Contact form specification

Four fields, three required, validated by one zod schema in `shared/` that runs in the browser for instant feedback and again on the server as the real gate.

Field	Input	Required	Rule	Error message
Name	`type="text"`, `autocomplete="name"`	Yes	Trimmed, 2 to 60 characters, Unicode letters, spaces, `.` `'` `-`	"Please enter your name." / "Name should be 2 to 60 letters."

Field	Input	Required	Rule	Error message
Email	`type="email"`, `autocomplete="email"`, `inputmode="email"`	Yes	Valid email format, max 254 characters, lower-cased	"Please enter your email." / "Please enter a valid email, like name@example.com."
Phone Number	`type="tel"`, `autocomplete="tel"`, `inputmode="tel"`	Yes	Strip spaces, dashes, brackets; optional `+91` or `0`, then 10 digits starting 6 to 9	"Please enter your phone number." / "Please enter a valid 10-digit mobile number."
Your Message	`<textarea>` 5 rows	No	Max 1000 characters, live counter	"Message can be up to 1000 characters."
`website` (hidden)	honeypot, `tabindex="-1"`, `aria-hidden`	-	Must be empty	none shown

Behaviour.

- Validate a field when it loses focus, then live as the user corrects it; on Submit, validate all and move focus to the first invalid field.
- Required fields show a visible "(required)" text, not only an asterisk, for screen readers and seniors.
- Errors sit under each field in `error-600` with an icon, linked by `aria-describedby`, and announced through `aria-live="polite"`.
- Submit button: label "Submit", `rose-600` background, full width on mobile, 56 px tall. While sending it shows a spinner, reads "Sending..." and is disabled to block double submits.
- Success: form is replaced by a confirmation card, "Thank you, {name}. We have received your message and will get back to you soon." with a "Send another message" link.

- Server 422: map `fields` back onto the matching inputs. 429: "Too many attempts, please try again in a few minutes or call us." Network or 500: keep the typed data and show the phone numbers as a fallback.
- Layout: two columns (form left, Reach us right) from 1024 px; stacked single column below, each field full width (requirement K5).
- A one-line note under the button links to the Privacy Policy: "We use your details only to respond to you."

13. Responsiveness, accessibility and senior-friendly UX

Target is WCAG 2.2 level AA on every page, designed at 360 px first and tested up to 1440 px.

Responsive rules.

Element	Mobile (< 768 px)	Tablet (768 to 1023 px)	Desktop (1024 px +)
Header	Logo, call icon, hamburger	Same as mobile	Full nav + Fix an Appointment
Hero	Text then image, buttons stacked full width	Text then image	Text left, image right
Why Choose / Amenities	1 column / 2 columns	2 columns / 3 columns	4 columns / 5 across
Treatment cards	1 column	2 columns	3 columns
Testimonials	1 card, carousel	1 card, carousel	3 cards side by side
Doctor grid	1 column	2 columns	3 columns
Contact	Form, then Reach us	Form, then Reach us	Two columns
Footer	4 stacked blocks	2 x 2 grid	4 columns
Sticky action bar	Shown	Hidden	Hidden

Accessibility checklist.

- [ ] Semantic landmarks (`header` , `nav` , `main` , `footer`), one `h1` per page, no skipped heading levels.
- [ ] "Skip to main content" link as the first focusable element.
- [ ] Text contrast at least 4.5:1, UI borders and icons at least 3:1 (tokens in section 6 already pass).
- [ ] Visible 3 px `lavender-700` focus ring on every interactive element; full keyboard use including menu, carousel and accordion.
- [ ] Tap targets at least 48 x 48 px with 8 px spacing.
- [ ] Carousel: pause button, previous/next buttons with labels, no auto-rotate under reduced motion (WCAG 2.2.2).
- [ ] Accordion: `button` headers with `aria-expanded` and `aria-controls` .
- [ ] Mobile menu traps focus while open, closes on Escape, returns focus to the hamburger.
- [ ] Every image has meaningful `alt` ; decorative ones use `alt=""` .
- [ ] Form labels always visible (no placeholder-only labels), errors announced.
- [ ] Page usable at 200% browser zoom and 320 px width with no horizontal scroll.
- [ ] `lang="en"` on `<html>` ; page `<title>` unique per route.

For senior citizens and anxious patients.

- 18 px body text, generous line height, short paragraphs.
- Phone numbers are always tappable `tel:` links; the call button sits next to Fix an Appointment.
- Plain labels ("Call the clinic", not icons alone); icons always paired with text.
- No pop-ups, no auto-playing video or sound, no clinical imagery that could alarm children.
- Hours and address visible in the footer of every page, so nobody has to hunt for them.

14. SEO and performance

Release gate: Lighthouse mobile scores of at least 90 Performance and 100 for Accessibility, Best Practices and SEO on every route, checked in CI.

SEO

- **Prerendered HTML** for all 6 content routes, so crawlers and WhatsApp/Facebook link previews see full text.

- **Per-page meta** via each route's `meta` export: unique `<title>` (under 60 characters) and description (under 155), canonical URL, Open Graph and Twitter card tags, one 1200 x 630 share image.

Route	Title
`/`	Sakthi Dental Clinic, Hosur: Dental Care for Women, Children & Families
`/about`	About Dr. Anupriya & Our Dental Team, Sakthi Dental Clinic Hosur
`/treatments`	Dental Treatments in Hosur: Implants, Braces, Root Canal & More
`/faqs`	Dental FAQs, Sakthi Dental Clinic Hosur
`/contact`	Contact & Directions, Sakthi Dental Clinic, Anthivadi, Hosur
`/privacy-policy`	Privacy Policy, Sakthi Dental Clinic

- **Structured data (JSON-LD):** `Dentist` on every page (name, address, phones, email, opening hours, `medicalSpecialty`, `sameAs` for socials once given); `FAQPage` on `/faqs`; `BreadcrumbList` on inner pages. Validate with Google's Rich Results Test; rich display is not guaranteed.
- **Crawl files:** `sitemap.xml` generated at build from the route list; `robots.txt` allowing all and pointing to the sitemap; 404 page set to `noindex`.
- **Content structure:** one keyword-led `h1` per page ("dentist in Hosur", treatment names), descriptive link text (no "click here"), treatment anchors with readable slugs.
- **Local SEO handover note:** recommend the clinic claim or update its Google Business Profile with the same name, address and phone (NAP) as the site.

Performance budgets

Metric (mobile, 4G)	Budget
Largest Contentful Paint	under 2.5 s
Interaction to Next Paint	under 200 ms

Metric (mobile, 4G)	Budget
Cumulative Layout Shift	under 0.1
JavaScript per route (gzipped)	under 120 KB
Hero image	under 150 KB
Total page weight, Home	under 1 MB

How we hit them.

- Images: a `sharp` script converts every source image to AVIF and WebP at 480, 960 and 1600 px widths; `<picture>` with `srcset` and `sizes`; explicit `width` and `height` to prevent layout shift; `loading="lazy"` below the fold; hero gets `fetchpriority="high"` and a preload.
- Fonts: self-hosted, Latin subset, 2 families and 4 weights only, `font-display: swap`, preload the heading font.
- JS: route-level code splitting (automatic in React Router), no carousel or animation libraries, icons imported individually.
- Delivery: Brotli from Vercel, immutable cache headers on hashed assets, API responses cached 10 minutes.
- Third parties: none by default (no embedded map iframe, no chat widget), which also keeps the Privacy Policy's cookie section accurate.

15. Security and privacy

The only personal data the site handles is the contact form, so the plan protects that path end to end and keeps the site's behaviour consistent with the Privacy Policy we publish.

Threat or duty	Control
Spam and bots on the form	Honeypot field, 5 requests per 15 min per IP, 10 KB body limit
Injection (NoSQL, script)	zod whitelists fields and types; strip HTML; Mongoose `sanitizeFilter`; no user input in queries except validated values

Threat or duty	Control
Cross-site scripting on output	React escapes by default; no `dangerouslySetInnerHTML` except the Privacy Policy Markdown, rendered through a sanitizing Markdown renderer
Clickjacking, sniffing, mixed content	`helmet` with CSP (`default-src 'self'`, API origin in `connect-src`), `frame-ancestors 'none'`, HSTS; HTTPS everywhere
Cross-origin abuse	CORS allow-list only (production domain, Vercel previews, localhost)
Leaked secrets	`.env` git-ignored, secrets only in Vercel and Render dashboards, `npm audit` and Dependabot in CI
Database exposure	Atlas user with read/write on one database only, network access limited to Render's outbound IPs where the plan allows, daily Atlas backups
Over-collection	Only name, email, phone, message stored; IP stored as a salted hash, not raw
Data retention	12-month TTL on `contact_messages`, pending coordinator approval
Cookies	Site sets none by default; if analytics are added later, add a consent banner first
Children's privacy	Form asks no age or child data, matching section 6 of the policy
Error leakage	Generic messages in production, details only in server logs

The Privacy Policy text is published exactly as supplied; any change to it, including the India-law point in section 3, is the client's decision, not ours.

16. Testing and QA

Eight layers of testing: the first six run automatically in CI; real-device checks and content proofreading are manual passes on Days 8 and 9.

Layer	Tool	What it covers
Unit (shared)	Vitest	zod contact schema: valid input, each invalid field, phone formats (`9876543210`, `+91 98765 43210`, `09876543210`), 1001-char message
Component	Vitest + React Testing Library	ContactForm errors and focus, FaqAccordion keyboard and `aria-expanded`, TestimonialCarousel pause, AppointmentButton renders `data-cta` and calls `ctaHandlers.appointment`
API	Vitest + Supertest + mongodb-memory-server	Every GET returns seeded counts (15 treatments, 9 doctors, 14 FAQs, 3 testimonials, 5 amenities); POST 201, 422 per field, honeypot, 429 after 5 tries, mail failure still 201
End-to-end	Playwright on Chromium, Firefox, WebKit + Pixel 7 and iPhone 14 profiles	Nav links reach each page; Fix an Appointment visible on every page; View Full List goes to `/treatments`; footer links work; Home treatment cards jump to the right anchor; contact form happy path and error path; no horizontal scroll at 360 px
Accessibility	`@axe-core/playwright` on every route + manual keyboard and screen-reader pass (NVDA, VoiceOver iOS)	Zero serious or critical axe violations
Performance and SEO	Lighthouse CI on every PR	Budgets in section 14
Real devices	Android Chrome, iPhone Safari, Samsung Internet, desktop Chrome, Edge, Firefox, Safari	Layout, fonts, tap targets, `tel:` and `mailto:` links
Content proofreading	Side-by-side with the brief	Every heading and paragraph matches; Privacy Policy diffed character by character

Coverage targets: 90% on `shared/` and server services, 70% on client components. Bugs found in QA go into a GitHub issue with page, device, screenshot and severity.

17. Deployment, environments and CI/CD

Every pull request gets an automatic preview the coordinator can open on a phone, and `main` deploys to production only after lint, tests, build and Lighthouse all pass.

Environment	Client	API	Database	Trigger
Local	`localhost:5173` (Vite)	`localhost:4000`	Local MongoDB in Docker or Atlas dev cluster	`npm run dev` (both apps via `concurrently`)
Preview	Vercel preview URL per PR	Render staging service	Atlas `sakthi_staging`	Push to a PR
Production	Vercel, custom domain (expected `sakthidentalclinic.in`, to confirm)	Render web service	Atlas `sakthi_prod`	Merge to `main`

GitHub Actions pipeline (on every PR).

1. Install with `npm ci` (Node 24, cached).
2. `npm run lint` and `npm run typecheck` across workspaces.
3. `npm test` (unit, component, API against mongodb-memory-server).
4. `npm run build` (client prerender against the staging API).
5. Playwright E2E + axe against the built client.
6. Lighthouse CI on all 6 routes with the budgets from section 14.

Git workflow. `main` is protected; work happens on `feat/<area>` branches; Conventional Commits (`feat:`, `fix:`, `chore:`); one PR per feature with screenshots for UI changes; squash merge.

Go-live steps. Point DNS to Vercel, confirm HTTPS, set `CORS_ORIGINS` to the final domain, run `npm run seed` on production, submit `sitemap.xml` in Google Search Console, send one real test message through the form and confirm the clinic email arrives.

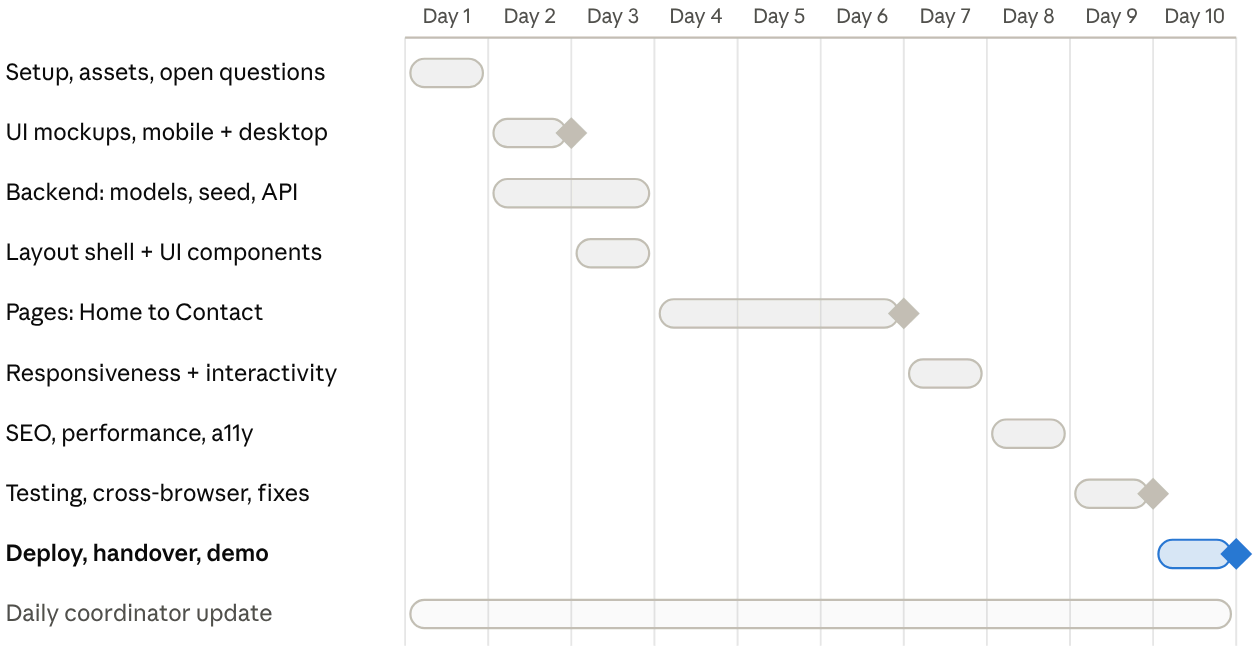
Handover package. README (setup, scripts, env vars, how to edit content in Atlas, where the CTA hook lives), `.env.example`, `CREDITS.md` for images and icons, and the final QA report.

18. 10-day timeline

The plan uses the full 10 days the brief allows, with every page feature-complete by Day 6 so Days 7 to 9 absorb feedback and bugs; if the coordinator wants 7 days, Days 7 and 8 merge and Day 9 becomes launch.

Feature-complete by Day 6, QA sign-off Day 9, launch Day 10

10 working days; each column is one day; diamonds mark coordinator sign-off gates



Gate 1, end of Day 2: coordinator approves the mockups
Gate 2, end of Day 6: every page feature-complete on the preview URL
Gate 3, end of Day 9: QA sign-off against section 19
Launch, Day 10: production live and handover delivered

10-day plan · 9 workstreams, 3 gates, launch on Day 10

The brief's four phases map directly: UI mockups (Days 1 to 2), responsiveness (Day 7), interactivity (Days 6 to 7), testing (Days 8 to 9).

Day-by-day deliverables.

1. **Day 1:** Send section 3 questions; download Drive assets; create repo, workspaces, ESLint/Prettier, CI skeleton; design tokens in code; seed `content.json` typed from the brief.

2. **Day 2:** Mobile and desktop mockups for all 6 pages (Figma); Mongoose models and seed script. Gate 1: coordinator approves mockups.
3. **Day 3:** All API endpoints with tests; Header, Footer, MobileMenu, MobileActionBar, Button, CTA components.
4. **Day 4:** Home page complete (8 sections) wired to `/content/home`.
5. **Day 5:** About and Treatments pages, including 15 anchors.
6. **Day 6:** FAQs, Contact (form + API + email), Privacy Policy, 404. Gate 2: every page live on the preview URL.
7. **Day 7:** Responsive pass at 360, 768, 1024, 1440 px; carousel, accordion, sticky bar, scroll motion.
8. **Day 8:** Meta tags, JSON-LD, sitemap, prerender check; image pipeline; accessibility fixes; Lighthouse budgets green.
9. **Day 9:** Playwright across 3 engines, real devices, content proofreading, bug fixes. Gate 3: QA sign-off.
10. **Day 10:** Production deploy, go-live checks, README and handover, demo to coordinator.

End-of-day update to the coordinator (copy and fill).

```
Sakthi Dental Clinic | Day N of 10 | <date>
Done today: ...
Preview link: ...
Planned tomorrow: ...
Blockers / questions: ...
On track for Day 10 launch: Yes / At risk (reason)
```

19. Definition of done

The project is done when every row in section 2 is ticked here and the coordinator has signed off on the production URL.

Content

- [ ] Home: hero copy, both CTAs, assurance banner, 4 Why Choose cards, 6 core treatments + View Full List, 3 testimonials rotating, 5 facilities with icons, footer.
- [ ] About: Dr. Anupriya section with all 4 paragraphs, Vision & Mission, 9 doctors.
- [ ] Treatments: all 15 with images and working anchors.
- [ ] FAQs: all 14 in order.

- [ ] Contact: form left, Reach us right with 4 icons and bold timings.
- [ ] Privacy Policy identical to the brief.
- [ ] Every open question from section 3 answered or logged with the default used.

Function

- [ ] Fix an Appointment and Emergency buttons present, styled, and routed through `ctaConfig.ts` with no booking logic.
- [ ] Contact form validates required fields and email format on client and server, saves to MongoDB, emails the clinic.
- [ ] All nav, footer and anchor links work; 404 page works.

Quality

- [ ] Lighthouse mobile: Performance 90+, Accessibility, Best Practices and SEO 100 on all routes.
- [ ] Zero serious axe violations; keyboard-only walk-through passes.
- [ ] No horizontal scroll from 320 px to 1440 px; checked on real Android and iPhone.
- [ ] Works in Chrome, Edge, Firefox, Safari, Samsung Internet.
- [ ] All CI checks green; no secrets in the repo.
- [ ] `CREDITS.md` lists every image and icon source with its licence.

Handover

- [ ] README, `.env.example`, seed script and QA report delivered.
- [ ] Production URL shared with coordinator; sitemap submitted.